

Project 2: Autonomous Mobile Assistant Robot Using Vision-Language-Action Models

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Number of Students: 8

Objective

Recent advances in Vision–Language–Action (VLA) models have enabled autonomous robots to perceive their surroundings, understand natural language instructions, and execute complex tasks through end-to-end multimodal reasoning. By integrating visual perception, language understanding, navigation, and robotic control, VLA models provide a unified framework for developing intelligent mobile service robots capable of operating in dynamic indoor environments.

The objective of this project is to design, develop, and implement an autonomous mobile assistant robot that can navigate indoor environments, interpret natural language commands, recognize and interact with objects, and perform service-oriented tasks with minimal human intervention. The proposed system will combine visual perception, semantic scene understanding, autonomous navigation, object manipulation, and Vision–Language–Action models to execute complex tasks specified through spoken or written instructions.

The project will initially be developed and validated in a high-fidelity simulation environment, such as NVIDIA Isaac Sim, before deployment on a physical mobile robotic platform. The system will be evaluated based on navigation accuracy, task completion rate, object interaction success, robustness, inference efficiency, and its ability to generalize to previously unseen environments and tasks.

Example Tasks

The robot should be capable of understanding and autonomously executing instructions such as:

- "Go to the laboratory and bring the water bottle."
- "Find an empty chair."
- "Deliver this package to Room 204."
- "Inspect the laboratory equipment."
- "Navigate to the meeting room and report the number of people present."
- "Locate the nearest fire extinguisher."
- "Pick up the toolbox and place it on the workbench."

- **"Guide a visitor to the Computer and Systems Engineering Department."**

Major Topics Covered

- 1. Robotics**
- 2. Mobile robot platform**
- 3. Differential-drive or omnidirectional locomotion**
- 4. Autonomous navigation**
- 5. Simultaneous Localization and Mapping (SLAM)**
- 6. Path planning and obstacle avoidance**
- 7. Robot control**
- 8. Sensor fusion**
- 9. ROS2 integration**
- 10. Computer Vision**
- 11. Object detection**
- 12. Semantic segmentation**
- 13. Scene understanding**
- 14. Human detection and tracking**
- 15. Depth perception**
- 16. Visual localization**
- 17. Artificial Intelligence**
- 18. Vision–Language–Action (VLA) models**
- 19. Vision-Language Models (VLMs)**
- 20. Natural language understanding**
- 21. Task planning**
- 22. Semantic reasoning**
- 23. Decision making**
- 24. Closed-loop visual feedback**

Suggested Hardware

- **Mobile robot platform (TurtleBot4, JetBot, JetRacer, or custom platform)**
- **NVIDIA Jetson Orin Nano**
- **Intel RealSense D435 or Luxonis OAK-D camera**
- **LiDAR sensor**
- **IMU**
- **Wheel encoders**
- **Microphone and speaker**
- **Robotic gripper (optional)**

- **Software**
- **ROS2**
- **Navigation2 (Nav2)**
- **NVIDIA Isaac Sim**
- **Movelt2 (if manipulation is included)**
- **PyTorch**
- **OpenVLA or Pi0/OpenVLA-compatible models**
- **YOLOv11**
- **Segment Anything (SAM)**
- **TensorRT**

AI Components

- **Visual perception**
- **Scene understanding**
- **Natural language understanding**
- **Object detection and recognition**
- **Semantic mapping**
- **Autonomous navigation**
- **Action planning**
- **Task execution**
- **Human-robot interaction**
- **Closed-loop perception and control**

Expected Deliverables

- **Autonomous indoor mobile robot**
 - **Voice-controlled natural language interface**
 - **Autonomous navigation and obstacle avoidance**
 - **Object search and recognition**
 - **Vision–Language–Action-based task execution**
 - **Digital twin developed in NVIDIA Isaac Sim**
 - **Real-world deployment and validation**
 - **Performance evaluation and benchmarking**
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